Part E — Written Justification

1, LOS histogram: asks how patient length of stay is distributed; a histogram suits a continuous measure and reveals right skew. Position along a common quantitative axis and binning encode frequency; bin width is **0.5 day**. **2, ward LOS boxplots:** asks whether LOS differs among wards; boxplots compare medians and spread across categories. Ward is encoded by x-position and LOS by a common y-position/scale, not colour alone; ward order is General Medicine, Cardiology, Day Surgery, Intensive Care. **3, systolic–diastolic scatter:** asks whether the two blood-pressure measures co-vary; a scatterplot exposes association and outliers. Both variables use quantitative position on labelled axes in **mmHg**; points use transparency rather than extra category colours. **4, correlation heatmap:** asks which cleaned measures are associated; a heatmap compactly compares pairwise Pearson correlations. A diverging palette is centred at **r = 0**, bounded **–1 to +1**, with every cell annotated numerically so colour is not the only channel. MAP is excluded because it is derived from systolic/diastolic pressure.

Parameters defended

Figure size **15×10 in** and `constrained_layout=True` keep four panels and labels legible. Histogram `binwidth=0.5` day balances detail and readability and is declared on the axis. Boxplot ward order is fixed for consistent comparison; `showfliers=False` is only a display choice, **not** data cleaning, and extreme LOS records are separately flagged/excluded under the stated >60-day rule. Scatter size **18** and alpha **0.35** reduce overplotting; title reports Pearson **r**. Heatmap uses Pearson correlation, `center=0`, `vmin=-1`, `vmax=1`, `annot=True`, `fmt='.2f'`, `square=True`, `linewidths=.5`, and 7-point annotations for comparability/readability. Palette comes from `vizlib.PALETTE`; top/right spines are removed; labels state units; source note names the input and cleaning rules. The named `fig` is saved **before** closing at **300 dpi** with `bbox_inches='tight'`.

Exclusions, changes and ethics

No whole patient rows were deleted; invalid values were set missing in the analysis copy and the original CSV was preserved. Mixed dates: **0** unparseable after parsing; ward labels: **6,527** normalized (not removed). Heart rate under Part D's predeclared plausibility rule (outside **30–220 bpm**): **37 values** set missing; mean moves **78.8331 → 78.0469 bpm**. For the required stricter device-fault statement, readings **above 250 bpm: n=29**; mean moves **78.8331 → 77.9843 bpm** when only those readings are excluded. BMI outside **10–80 kg/m²: 486 values** set missing; mean among valid entries moves **66.1770 → 27.3463 kg/m²**. LOS above **60 days: 18 values** set missing for the plotted distribution/boxplot; mean moves **2.3091 → 1.9454 days**. The >60-day threshold flags records for source verification; it does not prove every long stay is erroneous. Appendix histogram shows the heart-rate distribution before exclusion. The same physiologic/device plausibility rules would be applied regardless of whether exclusions strengthened or weakened the observed conclusion; otherwise the process would be outcome-driven editing, not cleaning.